

## Phase A — Immediate, Push-Delivered Updates with Granular Live Status

**Branch:** main · **Connector:** 1.2.10 · **Scope:** dashboard-triggered plugin / theme / WordPress-core / connector-self updates that **start within seconds** instead of waiting minutes for the next WP-Cron heartbeat.

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### 1. Problem

Previously an update requested from the dashboard was delivered **only by the pull channel**: the API set the command to pending, and the site picked it up on its next heartbeat. Heartbeats run from **WP-Cron, which only fires when the site receives a visitor**. On low-traffic sites a requested update could sit pending for many minutes with no visible progress, and the UI showed a single coarse “queued / in progress” state with no way to tell *why* nothing was happening or whether it had silently died.

### 2. What changed (summary)

Updates now use **two delivery channels** plus **granular lifecycle reporting** and **self-healing** so a command can never be stuck forever:

1. **Push channel (new, primary)**: on click the API immediately signs and POSTs the command straight to the site’s REST API. The connector validates it, acknowledges acceptance, and starts a background worker **without waiting for a visitor**. Typical start latency: seconds.
2. **Pull channel (fallback)**: if the site’s connector is older than 1.2.10, or the push request fails (unreachable, timeout, non-2xx), the command stays pending and is delivered on the next heartbeat exactly as before. Nothing regresses for old connectors.
3. **Granular status**: Queued → Starting → Downloading → Installing → Verifying → Completed, plus terminal Failed / Rolled Back — surfaced live in the dashboard with a step progress bar.
4. **Never stuck**: job locking, stale-lock recovery, duplicate-command protection, a server-side stale-command reconciler, and one-shot job tokens.

### 3. Files changed

#### API (apps/api)

| File                | Change                        |
|---------------------|-------------------------------|
| app/Models/Site.php | New granular status constants |

| File                                                                                    | Change                                                                                                                                                                                                                                                                                                |
|-----------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                         | (UPDATE_CMD_QUEUED/STARTING/DOWN LOADING/INSTALLING/VERIFYING/ROLLED _BACK); UPDATE_CMD_IN_FLIGHT / UPDATE_CMD_TERMINAL sets; UPDATE_CMD_STALE_MINUTES = 20; PUSH_UPDATE_MIN_VERSION = '1.2.10'; update_command_id fillable; supportsPushUpdate(), isUpdateInFlight(), reconcileStaleUpdateCommand(). |
| app/Services/Connector/ConnectorClient.php ( <i>new</i> )                               | Signs (HMAC) and POSTs the command to <home_url>/wp-json/marqira/v1/execute-update (with ?rest_route= fallback on 404). Returns {pushed, state, error, http}.                                                                                                                                         |
| app/Http/Controllers/Api/Dashboard/SiteController.php                                   | requestUpdate() reconciles stale commands, generates a command_id, attempts push when supported, records push outcome in the audit log, and returns a message reflecting push vs pull. updateStatus() reconciles stale before returning. Payload now carries command_id + in_flight.                  |
| app/Http/Controllers/Api/V1/UpdateCommandController.php                                 | Accepts the granular statuses + rolled_back; validates optional command_id and ignores acks whose id does not match the in-flight command (backward compatible); sets completed_at on any terminal status.                                                                                            |
| app/Http/Controllers/Api/V1/HeartbeatController.php                                     | Pull-path command now carries a stable command_id (generated once, persisted).                                                                                                                                                                                                                        |
| database/migrations/2024_01_07_000001_add_update_command_id_to_sites.php ( <i>new</i> ) | Adds nullable update_command_id (string 64) to sites. Reversible.                                                                                                                                                                                                                                     |

### Connector (wordpress/marqira-connector, v1.2.10)

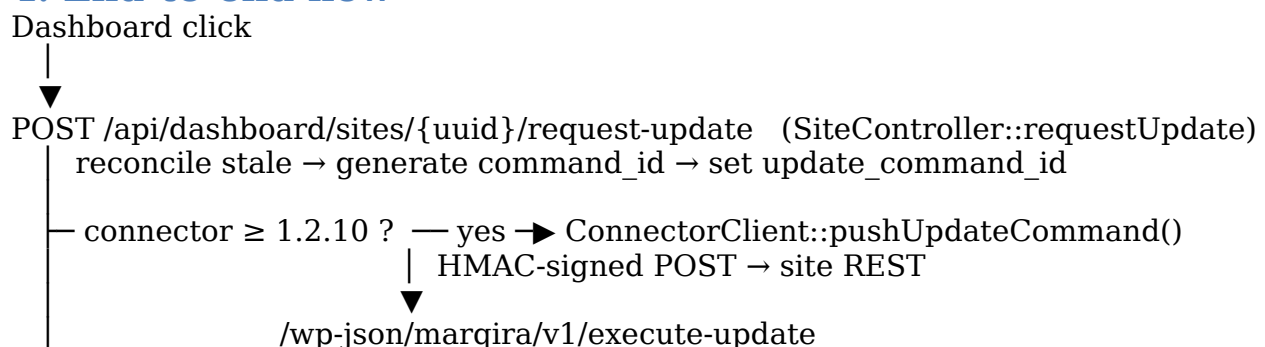
| File                                                  | Change                                                                                                                                                                                                                                                  |
|-------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| includes/class-marqira-hmac-server.php ( <i>new</i> ) | Inbound HMAC verifier for pushed requests: validates the 5 X-MarQira-* headers, site/kid match, $\pm 300s$ timestamp, constant-time signature over a canonical string bound to a fixed logical path, and nonce replay protection (transient, 600s TTL). |
| includes/class-marqira-rest-                          | Registers POST /marqira/v1/execute-update                                                                                                                                                                                                               |

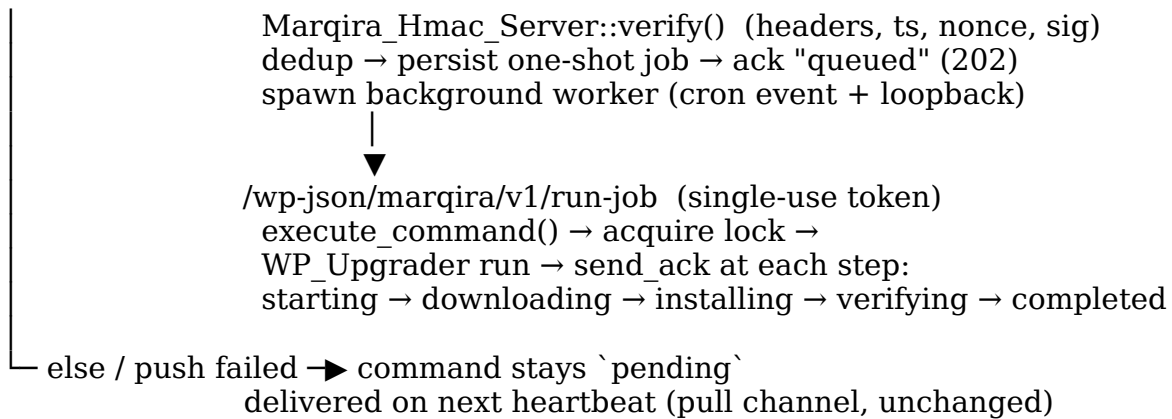
| File                                     | Change                                                                                                                                                                                                                                                                                                                   |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| controller.php ( <i>new</i> )            | (HMAC-verified) and POST /marqira/v1/run-job (single-use token). Dedups, persists a one-shot job, acknowledges queued, and spawns a background worker via a cron event <b>and</b> a non-blocking loopback request so work starts immediately. Returns 202 Accepted.                                                      |
| includes/class-marqira-remote-update.php | Routed through execute_command(\$type, \$target,\$command_id) shared by both channels. Adds up-front dedup, job locking with stale-lock recovery by age (LOCK_MAX_AGE = 900s), a bounded processed-command list (PROCESSED_CAP = 50), and granular send_ack() calls at each lifecycle step (now tagged with command_id). |
| marqira-connector.php                    | Loads the two new includes, initializes the REST controller, version → 1.2.10.                                                                                                                                                                                                                                           |

### Dashboard (apps/dashboard)

| File                        | Change                                                                                                                                                                                                                                                                  |
|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| src/types/index.ts          | UpdateCommandStatus extended with the granular + rolled_back states; SiteUpdateCommand gains command_id and in_flight.                                                                                                                                                  |
| src/pages/WebsiteDetail.tsx | New in-flight status set; a <b>step progress bar</b> (Queued→...→Completed); tone/label maps for every state; polls every <b>5s</b> while in flight (was 15s); shows the distinct push-vs-heartbeat waiting message and the clear failure/timeout message from the API. |

## 4. End-to-end flow





Every send\_ack → POST /api/v1/update-command/ack (HMAC) →  
UpdateCommandController::ack()

records granular status; ignores acks with no in-flight command or a mismatched command\_id.

Dashboard polls GET /update-status every 5s while in\_flight, rendering the step bar.

## 5. Reliability guarantees

- **Starts without a visitor** — push + non-blocking loopback worker; does not depend on organic traffic to fire WP-Cron.
- **Clear failure when it can't start** — if the site is unreachable or rejects the push, the API keeps the command pending, records the reason in the audit log, and the dashboard shows the fallback message (delivery on next heartbeat) rather than a fake "success".
- **Never stuck forever** — reconcileStaleUpdateCommand() flips any command older than **20 minutes** with no terminal ack to failed ("Update timed out; you can retry"), unblocking a fresh request. Runs on every status poll and before each new request.
- **No double-execution** — up-front duplicate-command protection (command\_id), a job lock with **stale-lock recovery by age** (15 min), and a **single-use job token** for the loopback worker.
- **Ack integrity** — acks are HMAC-verified and ignored if they reference a command that is not the one currently in flight (command\_id mismatch), preventing a late ack from a superseded command from corrupting state.
- **No optimistic UI** — the dashboard only shows states the site actually reported; "Completed" appears only after the connector confirms it.

## 6. Testing performed

This VM cannot run a live WordPress site or the full Laravel + Postgres + Redis stack end-to-end. Everything below was validated

with unit/feature tests, the connector PHP harness, `php -l`, and the dashboard build. Items that require a real site are listed in §7.

- **API — Pest:** full suite **264 passed** (898 assertions). New: `tests/Feature/Dashboard/SitePushUpdateTest.php` (6 — push accepted→queued with signed headers, failed push→pending fallback, old connector→no push, stale→ failed reconcile, fresh not reconciled, stale unblocks new request); appended 4 ack tests to `tests/Feature/RemoteUpdateCommandTest.php` (granular states non-terminal, rolled\_back terminal, matching/mismatched command\_id).
- **Connector — php tests/run.php: 105 passed / 4 pre-existing harness failures** (unrelated: missing `wp-admin/includes/plugin.php` and `get_permalink` in the stub harness — present before Phase A). New: `tests/test-hmac-server.php` (10 — valid verify, tampered body, wrong site, wrong kid, expired ts, path binding, replay first/second, missing headers) and `tests/test-remote-update-dedup.php` (10 — dedup + lock/stale-recovery).
- **Static:** `php -l` clean on every changed PHP file.
- **Dashboard:** `tsc --noEmit clean`; `npm run build` succeeds.

## 7. Requires live-site verification (cannot be done in this VM)

The following behaviors are implemented and unit/feature-tested but must be confirmed against at least one real WordPress site (ideally WP 6.x and WP 7.1):

1. Push request reaches the site's REST endpoint through the site's real web server / firewall / Cloudflare, and the HMAC verifies end-to-end.
2. The non-blocking loopback worker actually spawns and runs the upgrader on the host (some hosts block loopback requests; the cron-event path is the backup).
3. WP\_Upgrader completes real plugin / theme / core / connector-self updates and each granular ack (downloading/installing/verifying) is emitted in order.
4. Stale-lock recovery under a genuinely crashed mid-update process.
5. Behavior on a host with `DISABLE_WP_CRON` / alternate cron.

## 8. Coolify deployment steps

No new environment variables are required. Redis is already required by the API (nonce manager) — no change there.

1. **Pull main** in the API and dashboard services (or trigger redeploy).
2. **API — run the new migration** (one new nullable column, safe/reversible):

```
php artisan migrate --force
```

In Coolify: API service → Terminal (or a post-deploy command) → `php artisan migrate --force`.

3. **API — clear/rebuild caches** if you cache config/routes:

```
php artisan config:clear && php artisan route:clear
```

4. **Dashboard — rebuild** (Coolify runs `npm run build` on deploy; confirm the build succeeds — it does locally).
5. **Publish the connector release:** upload releases/marqira-connector-1.2.10.zip via the dashboard's Plugin Releases screen and **activate** it, so sites can self-update to 1.2.10. Until a site is on 1.2.10 it keeps using the pull (heartbeat) channel — no breakage.
6. **No worker/queue change** is needed for Phase A (push is synchronous from the request; the background execution happens inside WordPress on the site).

## 9. Rollback

- Revert the deploy to the previous commit.
- The migration is reversible: `php artisan migrate:rollback` drops `update_command_id`. The API tolerates its absence (all reads are guarded).
- Sites already upgraded to connector 1.2.10 continue to function; the push endpoint simply goes unused if the API no longer calls it.